

# Things You Wanted to Know!

**Q.** I want a remote-controlled circuit for controlling the direction of a DC motor. I need to control the direction with the help of two switches. I also need to control the duration of running.

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**A.** The circuit for controlling a DC motor described here is in two

parts: a radio frequency (RF) transmitter unit and an RF receiver unit.

The circuit of the RF transmitter unit is shown in Fig. 1. The circuit is built around 5V voltage regulator 7805 (IC1), microcontroller (MCU) ATmega16 (IC2), 433MHz transmitter module TX1 and a few other components. The clock for the MCU is provided by an 8MHz crystal. The circuit is powered by 5V, which is provided by IC1. Two push-button switches S1 and S2 are used to send commands to the receiver unit to drive DC motor (M1) in clockwise or

anticlockwise direction, respectively. The transmitter (TX1) and receiver (RX1) work at 433MHz.

The RF based control proves to be more advantageous as compared to the infrared (IR) based control that limits the operating range to only a few metres.

The circuit of the RF receiver unit is shown in Fig. 2. It is built around 5V voltage regulator 7805 (IC3), MCU ATmega16 (IC4), H bridge driver 293D (IC5), 433MHz receiver module (RX1), 5V DC motor (M1) and a few other components.

The DC motor is controlled by the transmission of RF signals from the transmitter unit. The receiver receives these signals, which are interpreted by the MCU for direction control of the motor. Depending upon the switch being pressed at the transmitter unit, the program in the MCU sends the input to IC5 for driving the motor in the desired direction. The duration for motor movement is controlled by an inbuilt timer of the MCU. For example, 10-second time duration has been provided in the software, which will be forwarded to you through email.

The program is written in C language. You require Atmel Studio 6.2, which is the integrated development platform (IDP), to compile the code, PROGISP (version 1.72) to download the program and AVR-PRO+ V03 PCB as programmer.

Answers compiled by EFY joint director (training), Col. N.C. Pande (Retd). Letters and questions for publication may be addressed to Editor, Electronics For You, D-87/1, Okhla Industrial Area, Phase 1, New Delhi 110020 (e-mail: editsec@efy.in) and should include name and address of the sender

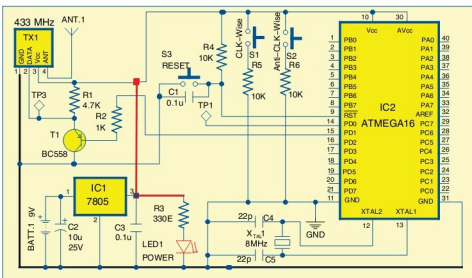


Fig. 1: RF transmitter unit circuit

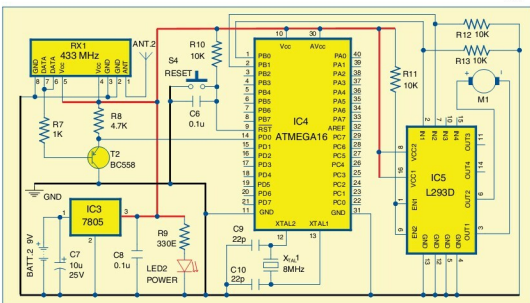


Fig. 2: RF receiver unit circuit